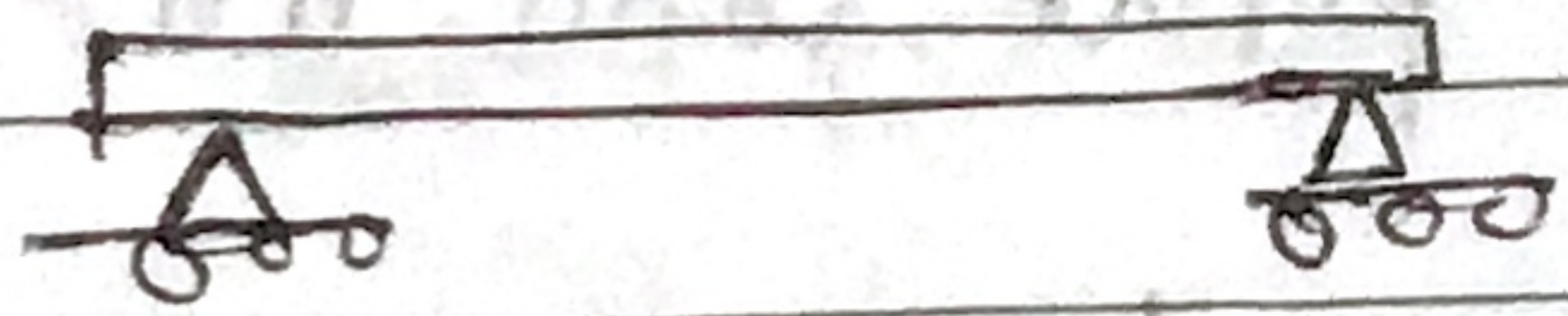


Types of Beams

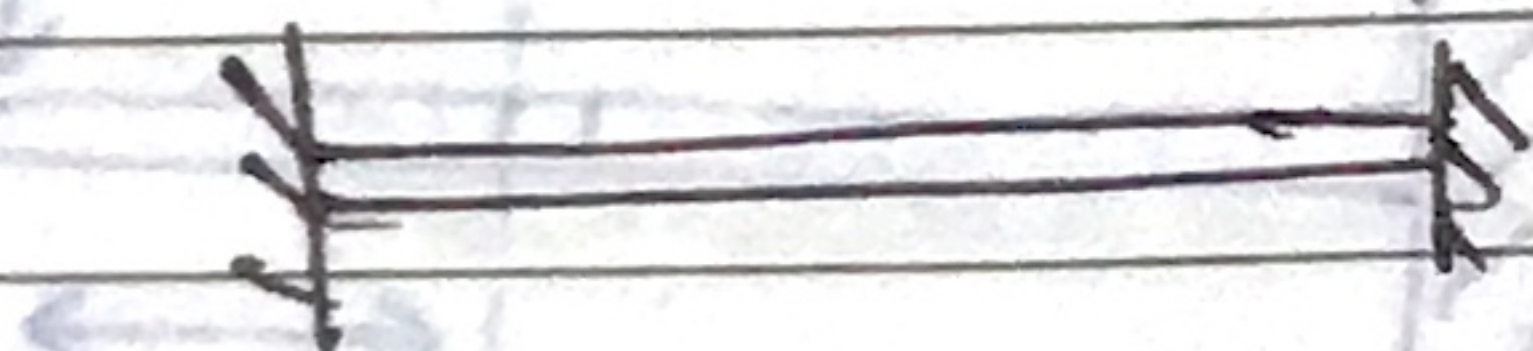
Simple supported



Cantilever



Fixed



Overhanging



Continuous



Statistically determinate

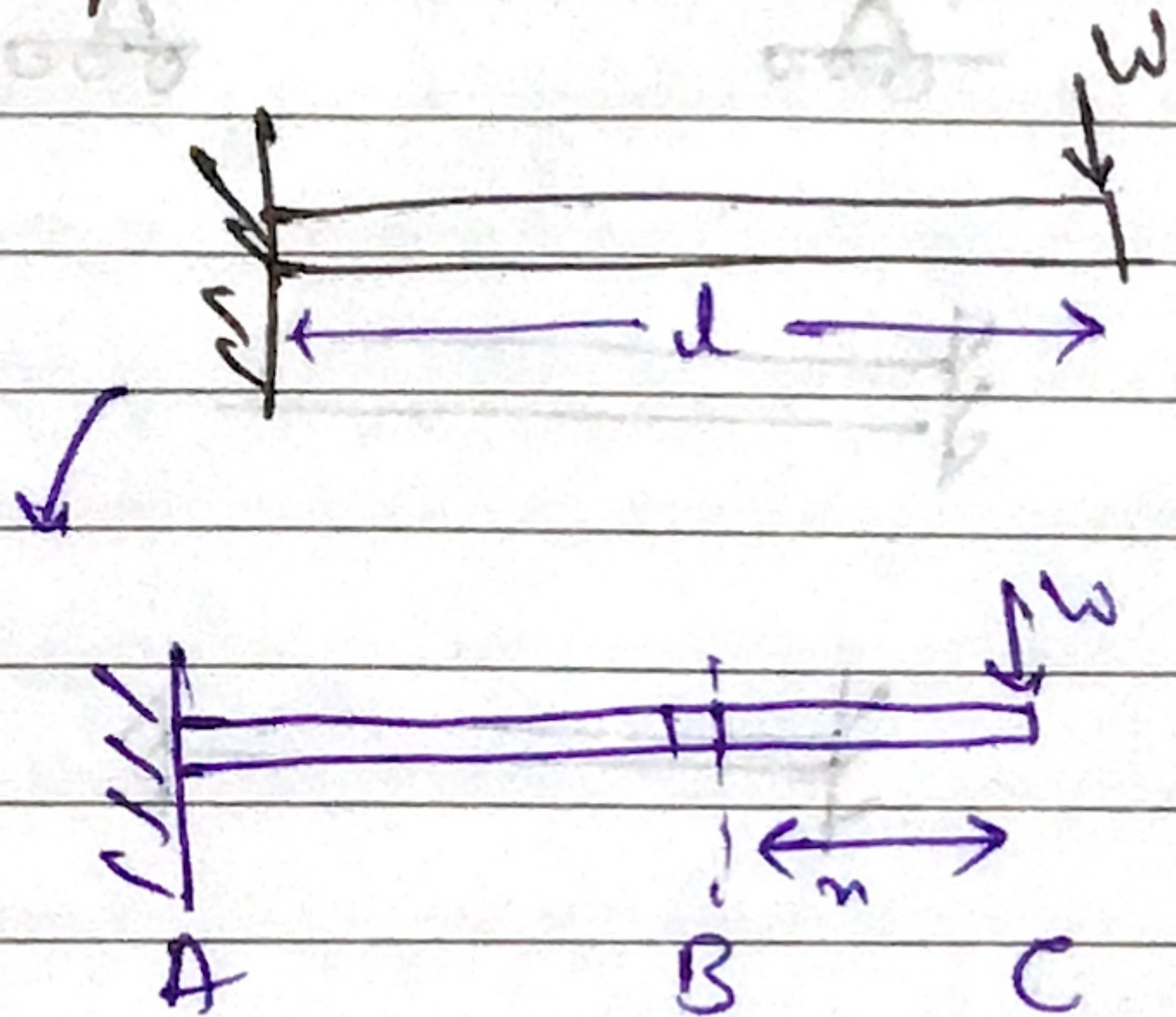
→ If the reactions can be found using only $\sum F_x = 0$, $\sum F_y = 0$, and $\sum M = 0$

Statistically indeterminate

→ If reactions could not be solved by $\sum F_x = 0$, $\sum F_y = 0$, $\sum M = 0$ and more relations are required such as deflection relation, slope deflection, etc.

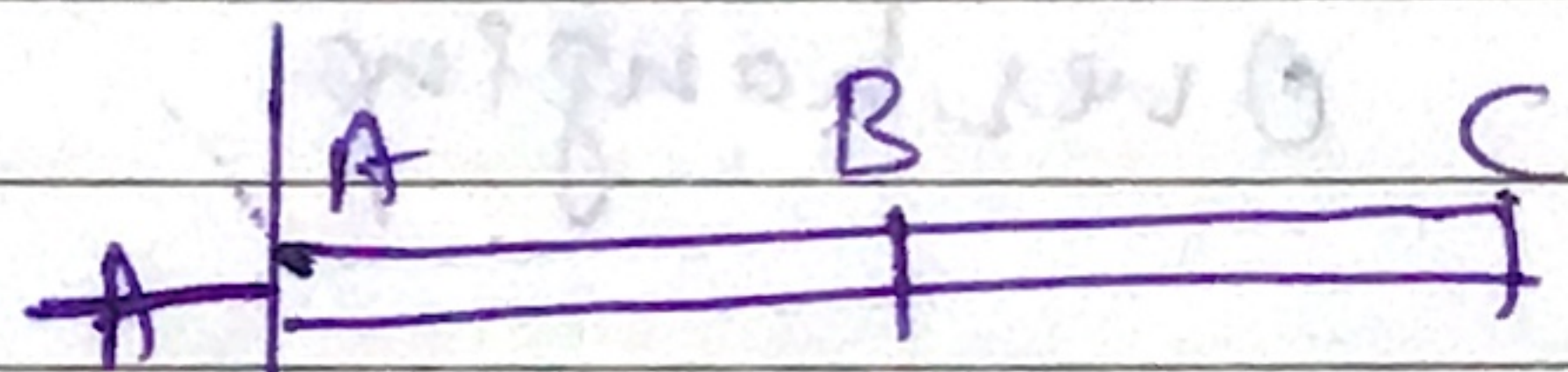
SFD and BMD

① point load on end of cantilever beam

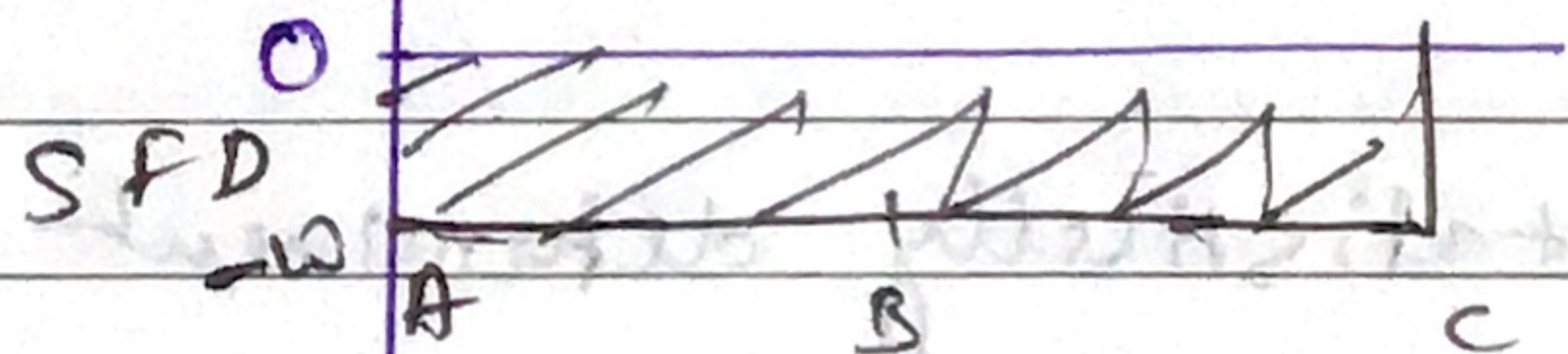


↓ (-ve)
clockwise

$$Sf_x = -W$$

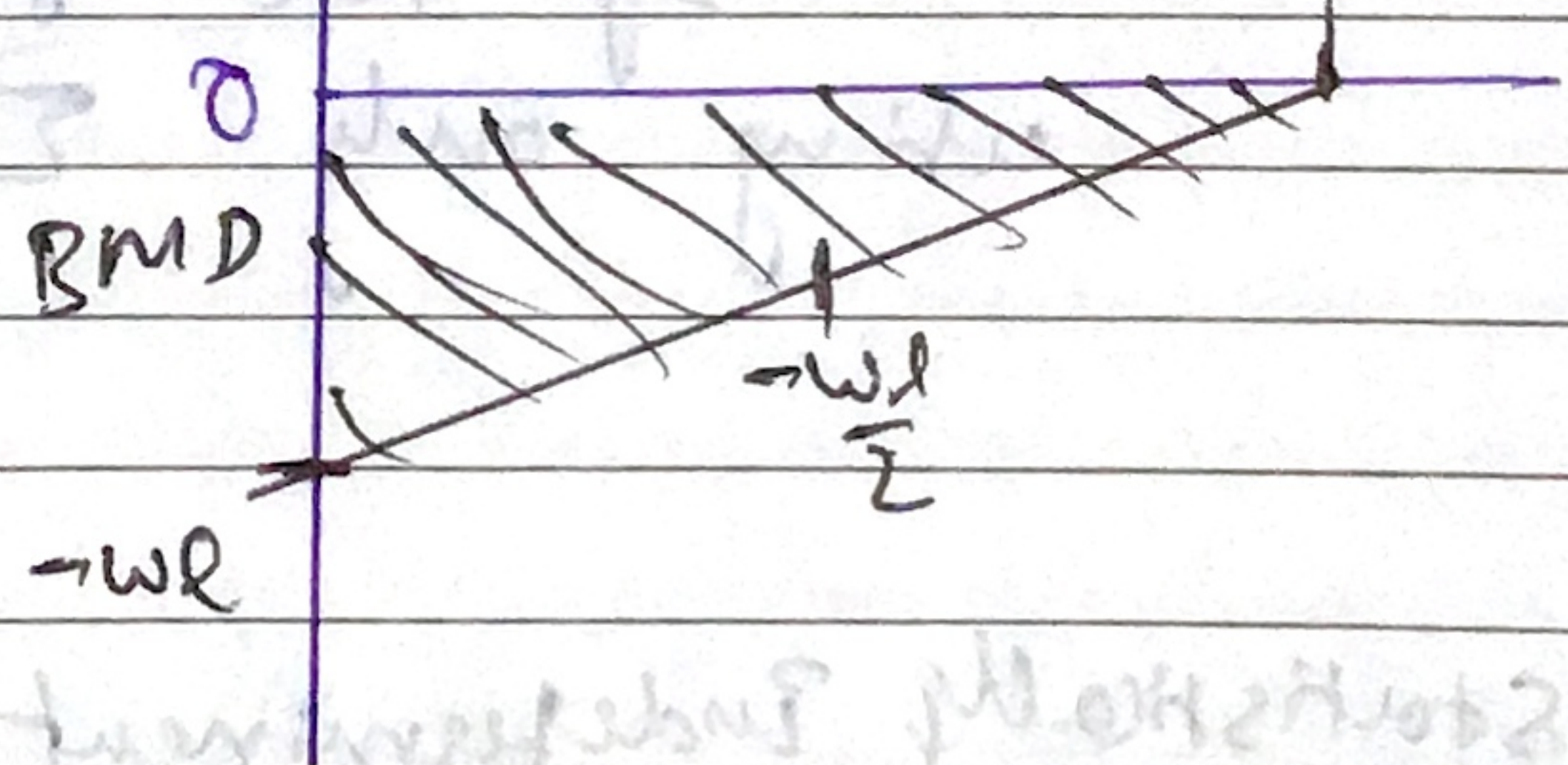


At $n=0$, $Sf_x = -W$
 $n=l/2$, $Sf_x = -W$
 $n=l$, $Sf_x = -W$

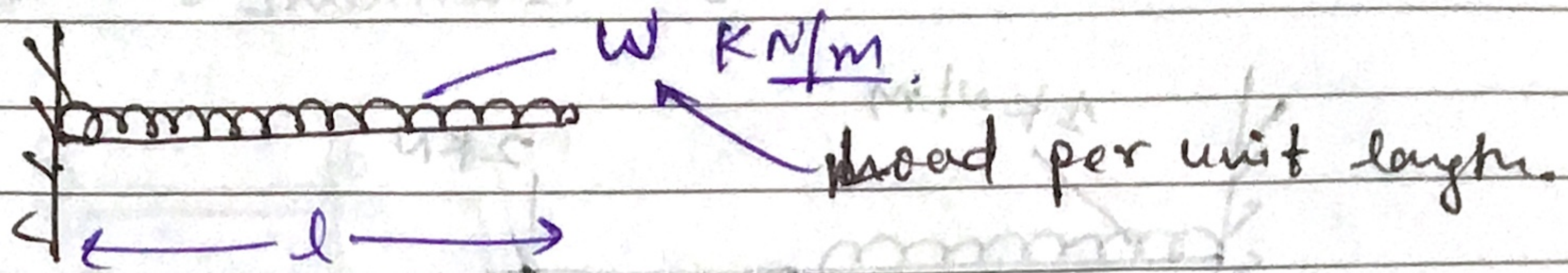


$$BM_x = -Wn$$

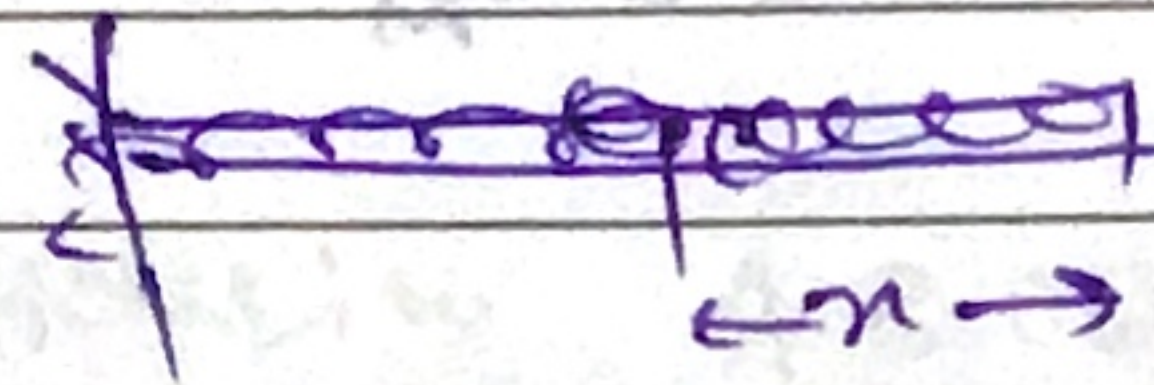
At $n=0$, $BM = 0$
 $n=l/2$, $BM = -Wl/2$
 $n=l$, $BM = -Wl$



② UDL on Cantilever beam



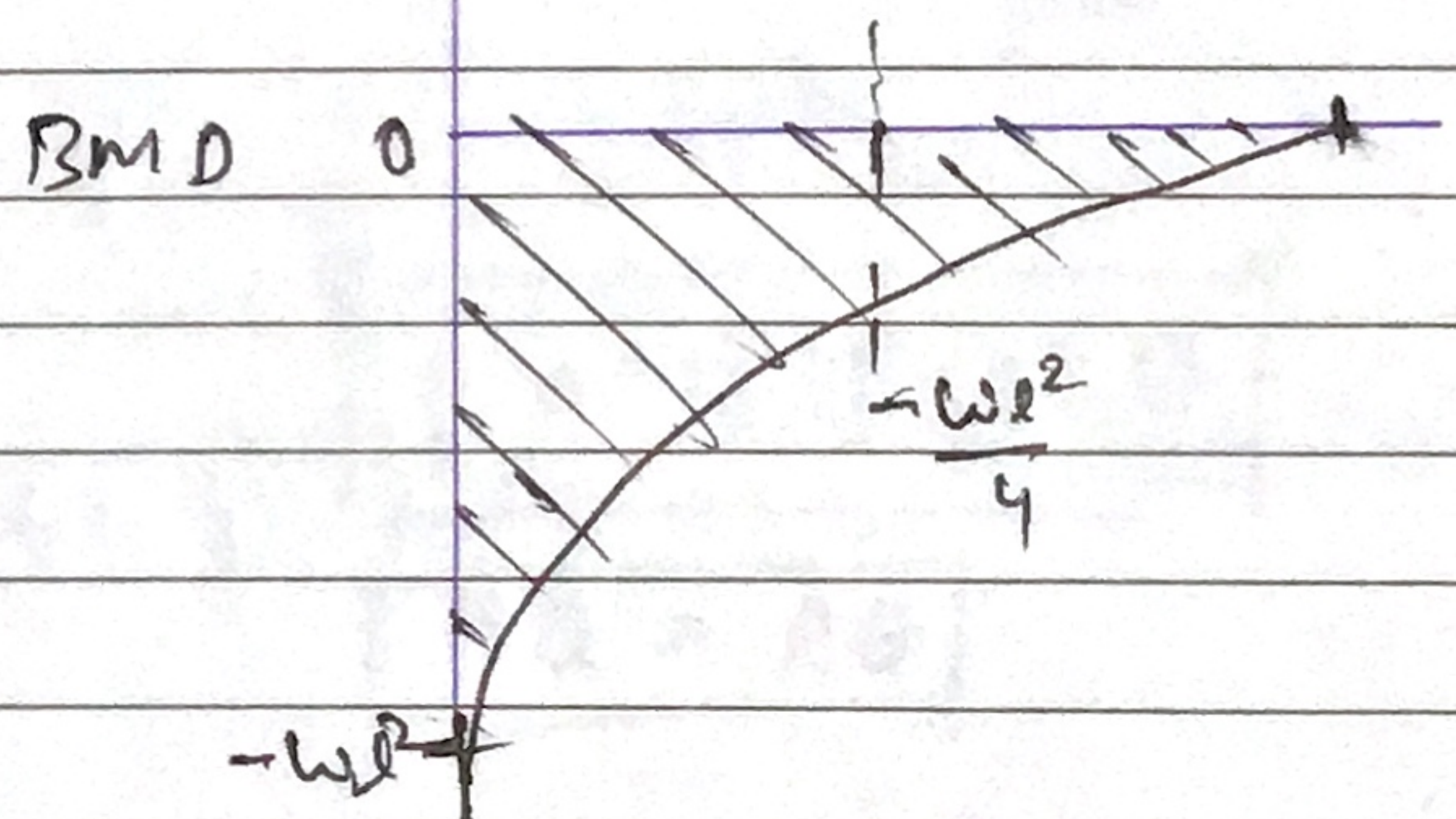
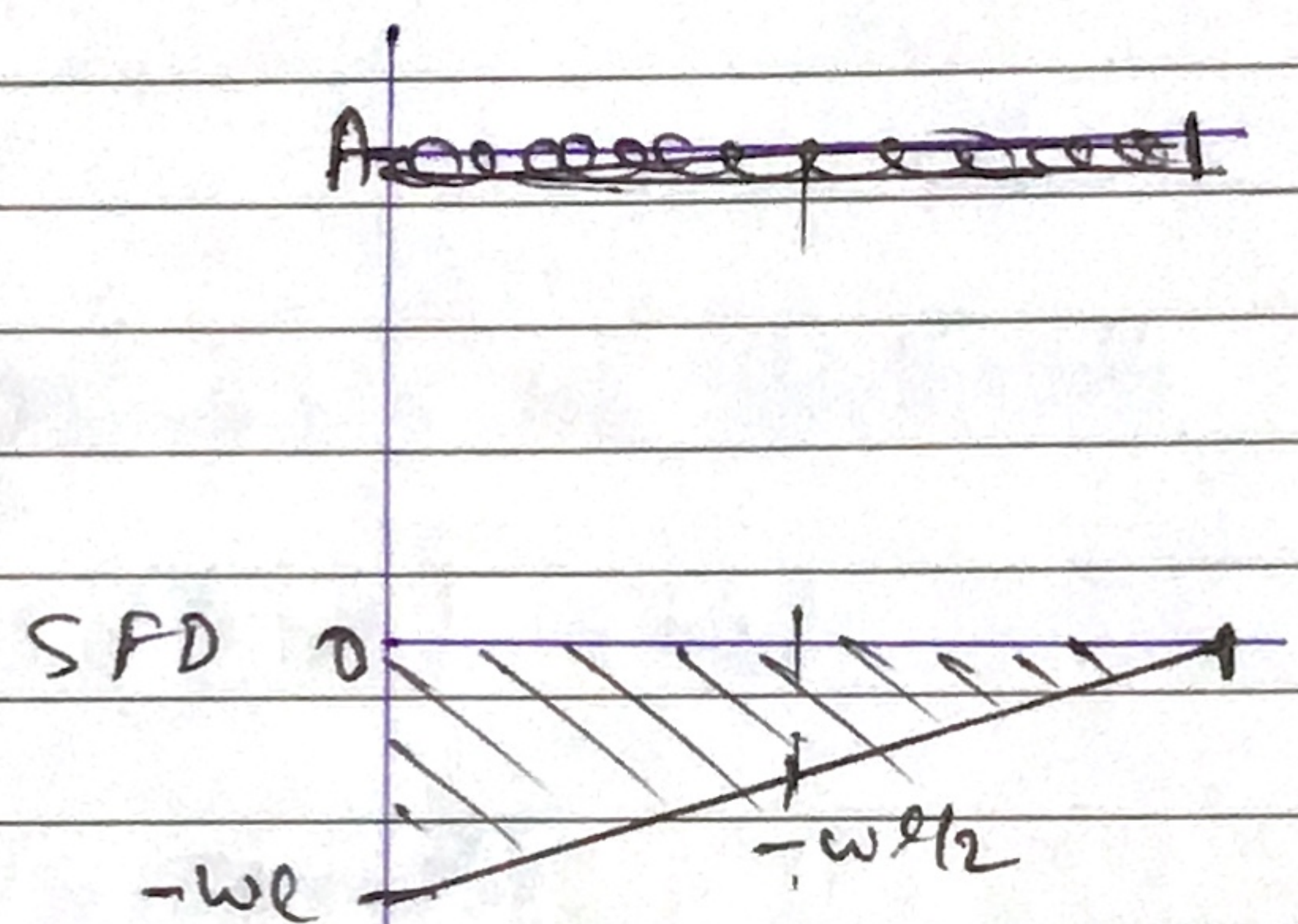
$$SF_x = -wn$$



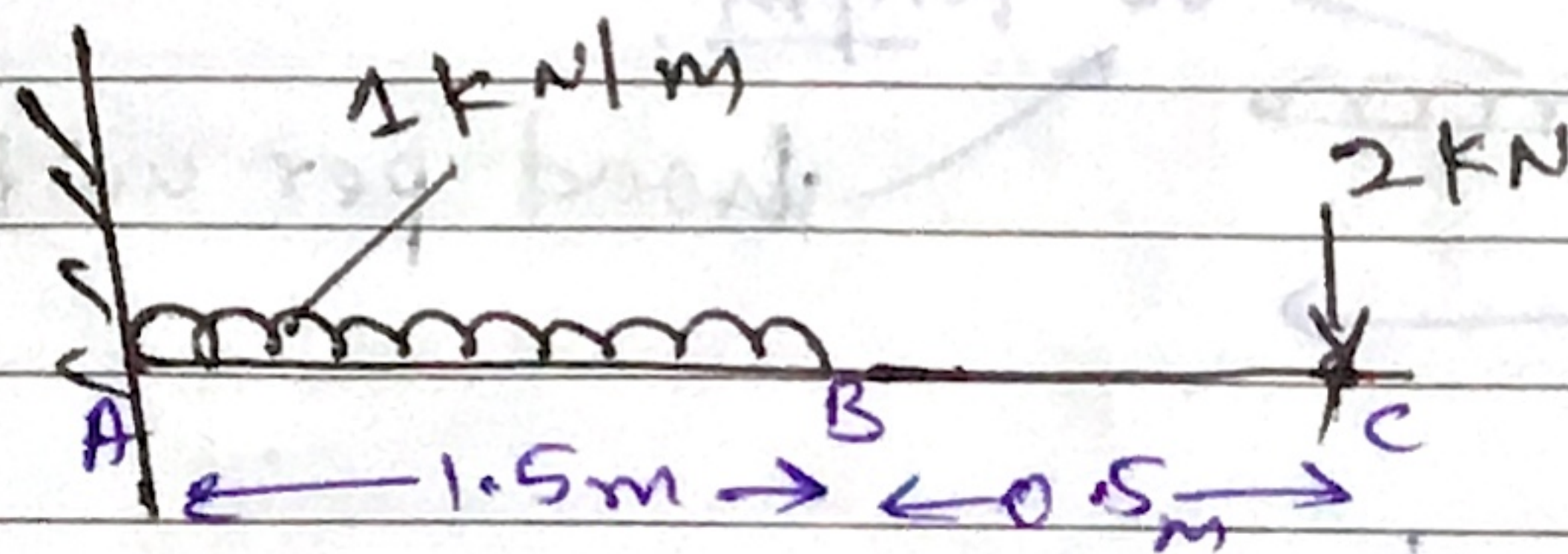
At $n=0$, $SF = 0$
 $n=l/2$, $SF = -wl/2$
 $n=l$, $SF = -wl$

$$BM = -wn^2$$

At $n=0$, $BM = 0$
 $n=l/2$, $BM = -wl^2/4$
 $n=l$, $BM = -wl^2$



③ Combination of UDL and point load on cantilever beam.



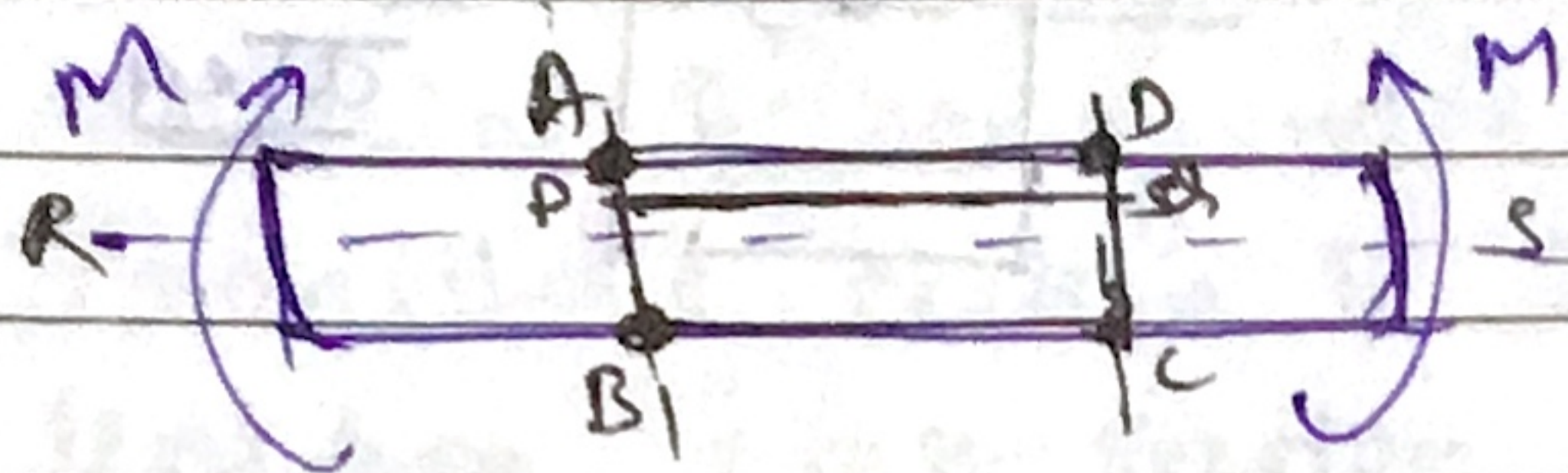
$$VW = 0.72$$

$$\begin{aligned} \text{At } A, V = 0, M = 0 \\ \text{At } B, V = 0.72, M = 0 \\ \text{At } C, V = 0.72, M = 0 \end{aligned}$$

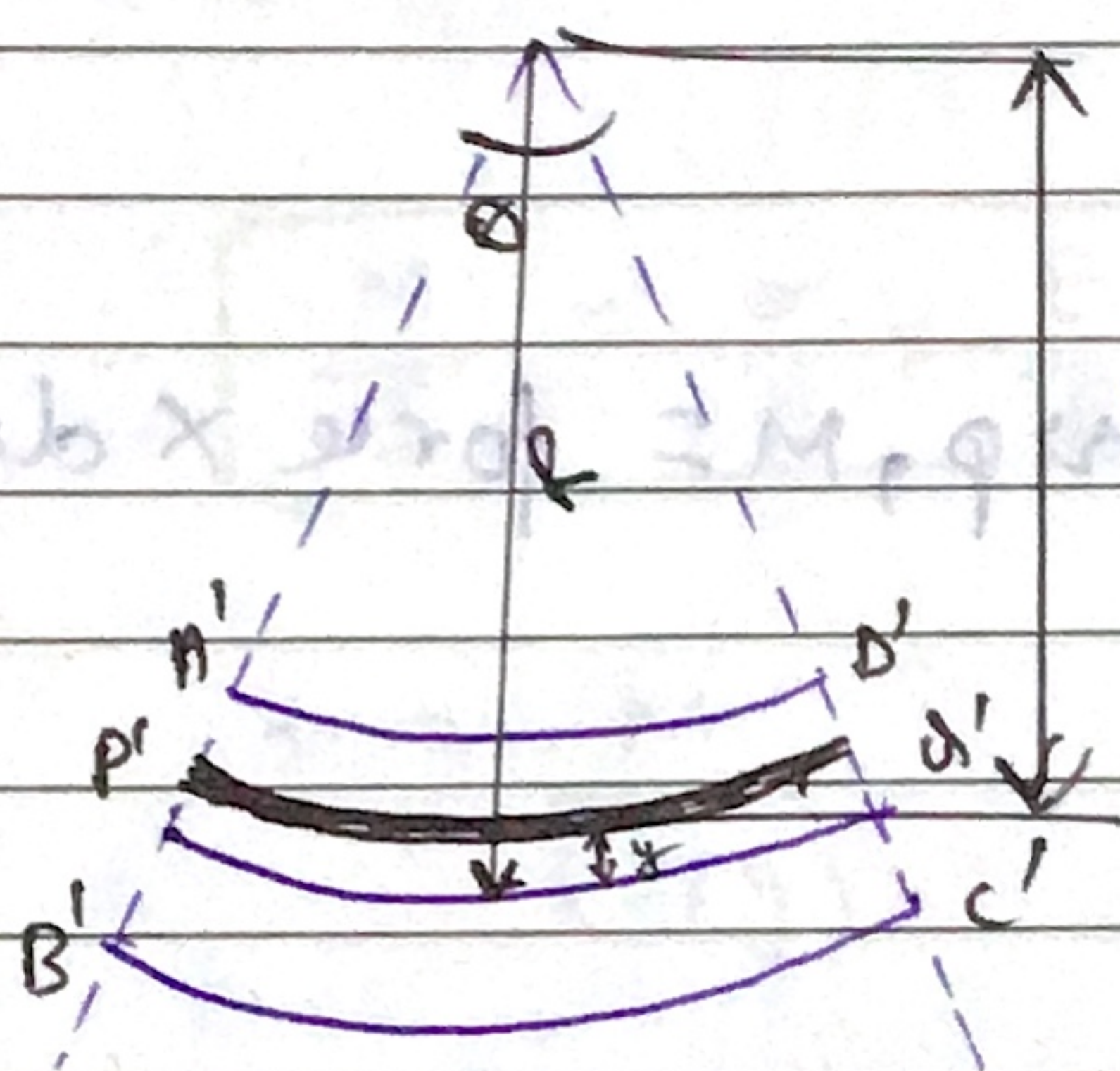
$$BM = -0.72$$

$$\begin{aligned} \text{At } A, BM = 0, V = 0 \\ \text{At } B, BM = -0.72, V = 0 \\ \text{At } C, BM = -0.72, V = 0 \end{aligned}$$

Bending Equation



Considering ABCD After bending



$$\epsilon = \frac{\Delta l}{l} = \frac{PQ - P'Q'}{PQ}$$

$$\epsilon = 1 - \frac{P'Q'}{PQ}$$

$$P'Q' = (R-y)\theta$$

$$\epsilon = 1 - \frac{(R-y)\theta}{R\theta}$$

$$PQ = R\theta$$

$$\epsilon = 1 - \left(1 - \frac{y}{R}\right)$$

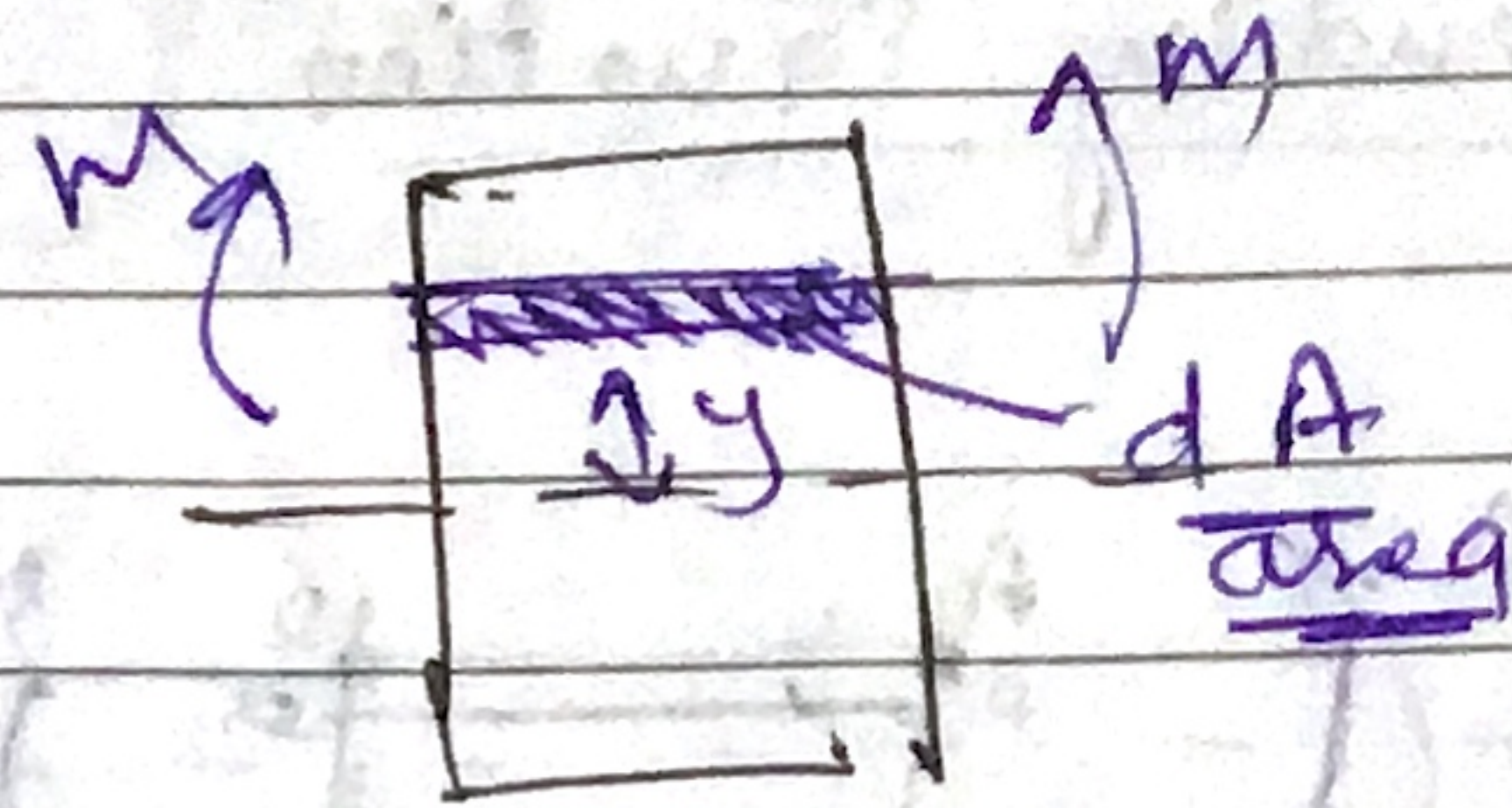
$$\frac{\sigma}{E} = \frac{y}{R}$$

or,

$$\frac{\sigma}{y} = \frac{E}{R}$$

$$\frac{\sigma}{y} = \frac{E}{R}$$

$$\sigma = \frac{E}{R} \cdot y$$



force on elementary strip $= \sigma dA$

moment on elementary strip, $M = \text{force} \times \text{distance} (y)$

$$dM = (\sigma dA) \times y$$

$$dM = \left(\frac{E}{R} \cdot y dA \right) y$$

$$dM = \frac{E}{R} y^2 dA$$

$$M = \frac{E}{R} \int y^2 dA$$

$$M = \frac{E}{R} \cdot I$$

$$\left[\frac{M}{I} = \frac{E}{R} \right] \#$$

$$\int y^2 dA = I$$

$$\frac{M}{I} = \frac{E}{R}$$

Now

$$\frac{M}{I} = \frac{E}{R} = \frac{\sigma}{y}$$

flexure formula

Assumptions

- ① Material is homogeneous and isotropic
- ② Transverse section is thin and flat.
- ③ Uniform cross-section
- ④ Bending within elastic limit.
- ⑤ Neutral axis is 11^r to face (upper and lower)
in undeformed condition.

$$\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$$

$$\sigma = \frac{M}{(I/y)}$$

$$\frac{I}{y} = Z$$

Section modulus

$$\sigma = \frac{M}{Z}$$

$$\sigma \propto \frac{1}{Z}$$

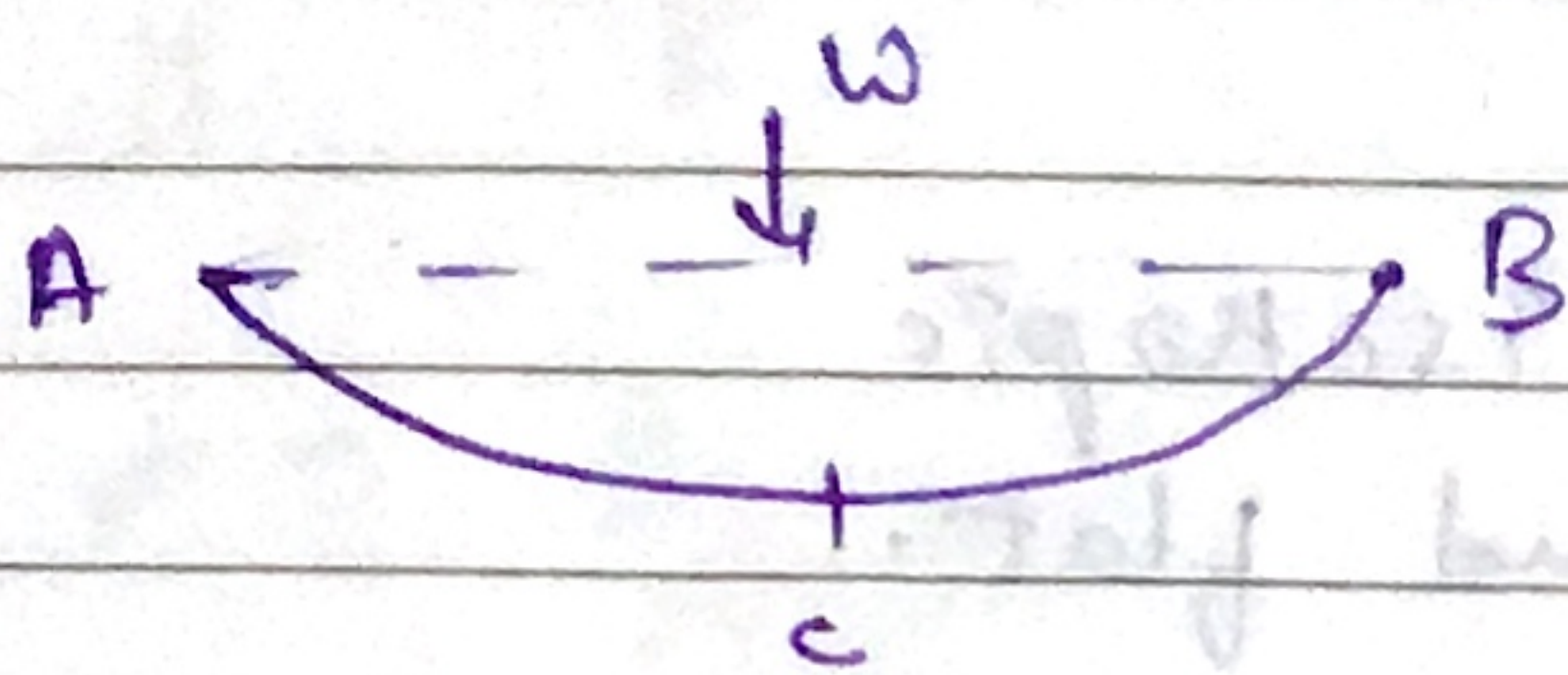
Refers to the ability of cross-section to resist deformation under Applied load.

Flexure Rigidity

$$\Rightarrow EI$$

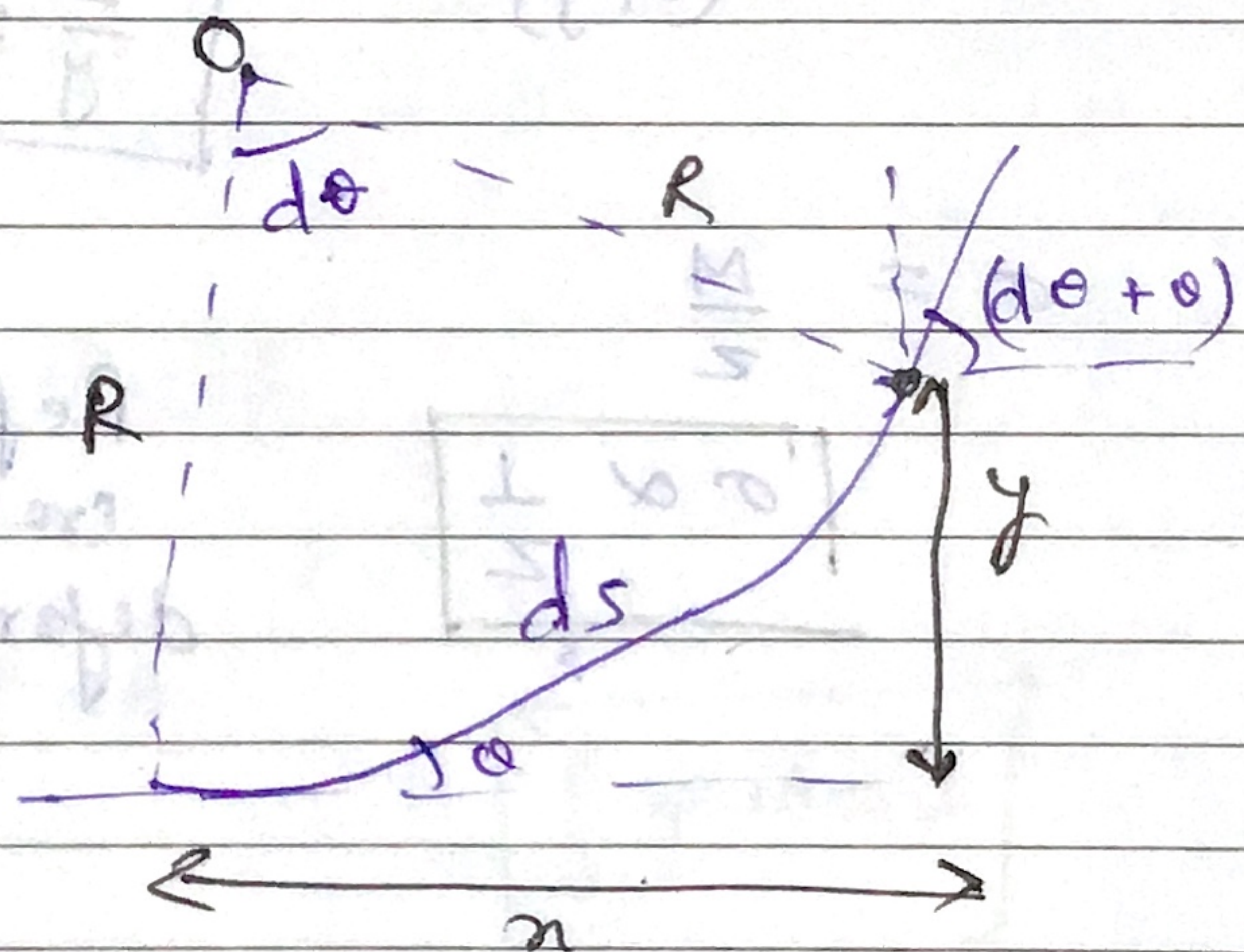
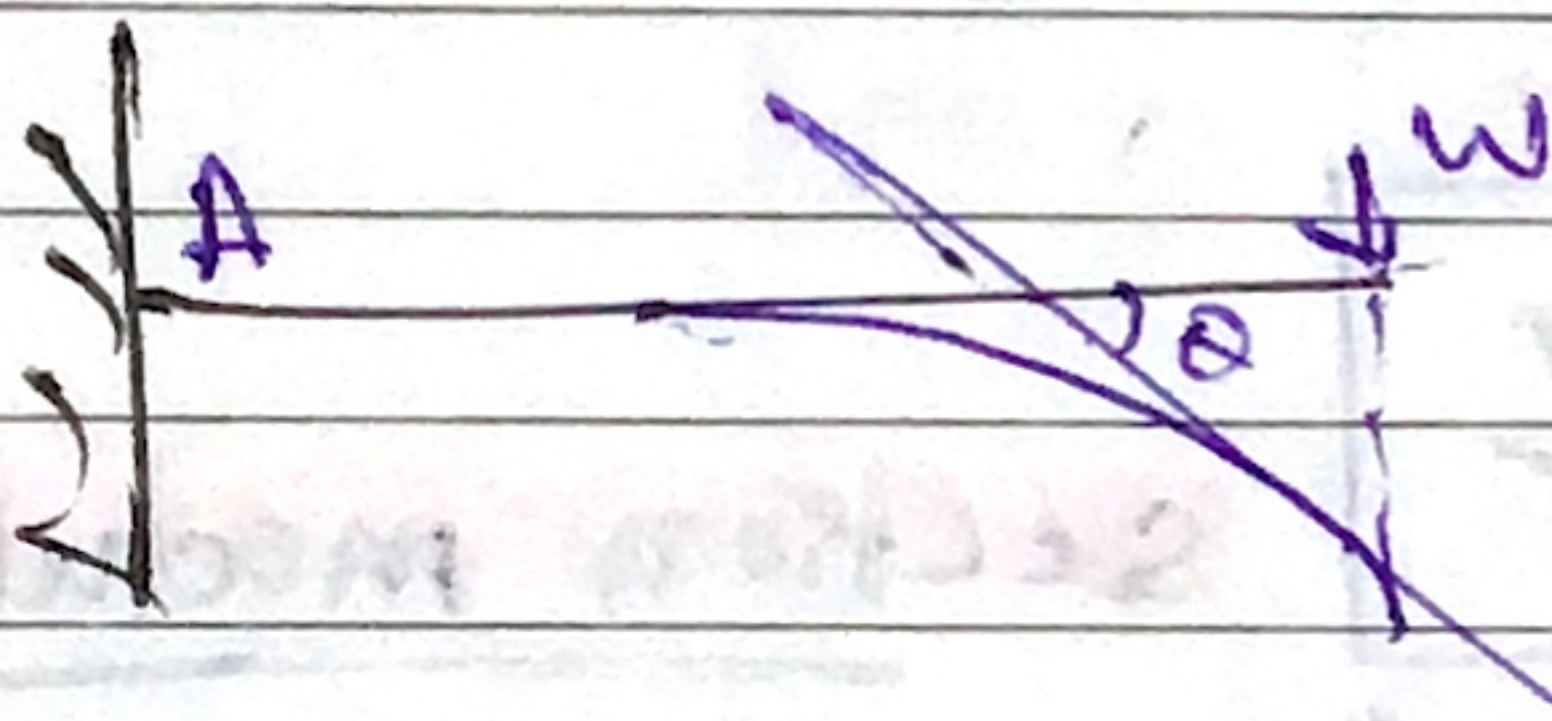
Beams ability to resist bending.

Deflection of Beam



At A θ_A
deflection = 0
At B θ_B
deflection = 0
At C $\theta = 0$
deflection = max

Consider a cantilever beam shown below



Arc will be formed after bending.

$$ds = R d\theta$$

$$\frac{1}{R} = \frac{d\theta}{ds}$$

$$\frac{ds}{dx} = \frac{dy}{dx} \tan \theta$$

$$\frac{dy}{dx} = \tan \theta$$

$$\theta = \frac{dy}{dx}$$

$$\frac{d\theta}{dx} = \frac{d^2y}{dx^2}$$

Good Write

for small θ $ds = dx$

$$\frac{1}{R} \frac{d\theta}{dx} = \frac{d^2y}{dx^2}$$

$$\frac{1}{R} = \frac{d\theta}{dx} = \frac{d^2y}{dx^2}$$

$$\boxed{\frac{1}{R} = \frac{d^2y}{dx^2}}$$

from flexure formula

$$\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$$

$$\frac{M}{EI} = \frac{1}{R}$$

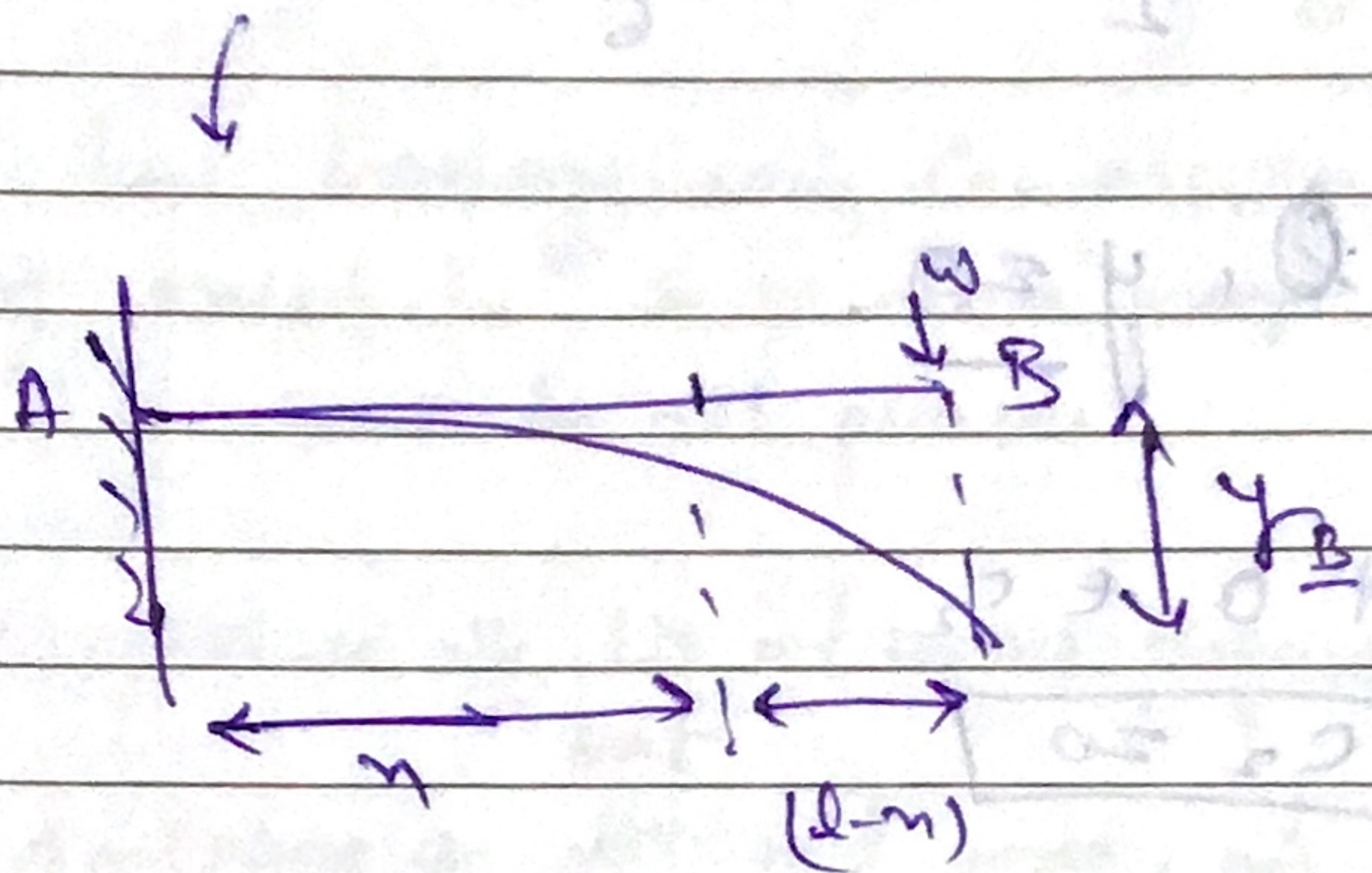
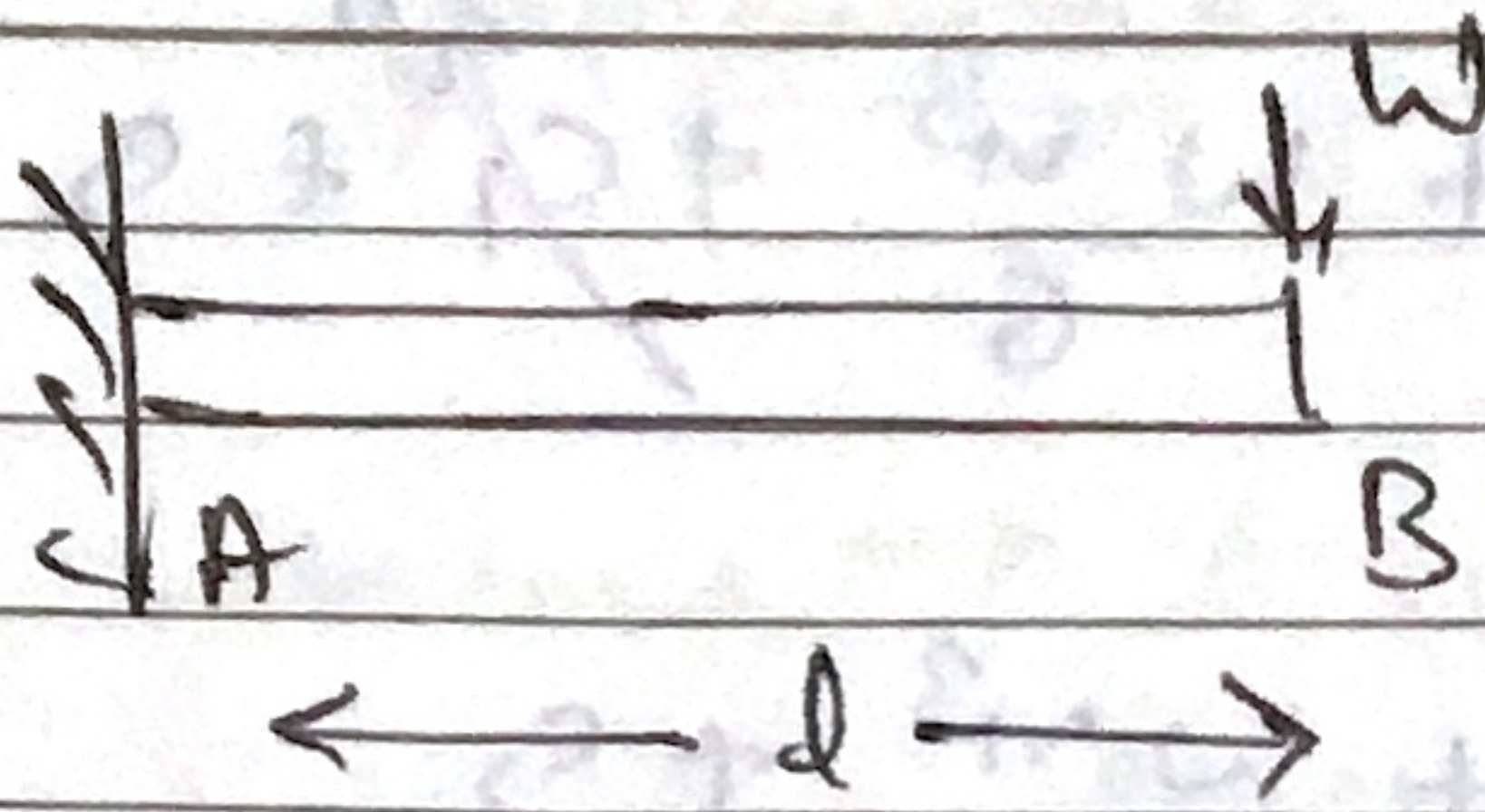
$$\boxed{\frac{M}{EI} = \frac{d^2y}{dx^2}}$$

$$\int \frac{dy}{dx} = \int \frac{M}{EI} dx$$

$$\boxed{y = \iint \frac{M}{EI} dx}$$

double integration

Q



$$M_x = -w(l-n)$$

$$= -wl + wn.$$

$$EI \frac{d^2y}{dn^2} = -wl + wn$$

$$EI \frac{dy}{dn} = -wn + w \frac{n^2}{2} + C_1$$

$$EI \frac{dy}{dn} = -wn + w \frac{n^2}{2} + C_1$$

At $n=0$, $\frac{dy}{dn} = 0$

$$0 = 0 + 0 + C_1$$

$$\boxed{C_1 = 0}$$

$$EI y = -w l \frac{x^2}{2} + w \frac{x^3}{6} + C_1 + C_2$$

$$EI y = -w l \frac{x^2}{2} + w \frac{x^3}{6} + C_2$$

for
At $x=0, y=0$

$$0 = 0 + 0 + C_2$$

$$\boxed{C_2 = 0}$$

$$EI y = -w l \frac{x^2}{2} + w \frac{x^3}{6}$$

for max^m, y , i.e. y at B (end)

$y_{\max}$ is at $x = l$

$$EI y = -w l \left(\frac{l^2}{2} \right) + w \frac{l^3}{6}$$

$$EI y = -w \frac{l^3}{2} + w \frac{l^3}{6}$$

$$= w l^3 \left(-\frac{1}{2} + \frac{1}{6} \right)$$

$$EI y = w l^3 \left(\frac{-6 + 2}{12} \right)$$

$$EI y = -w \frac{l^3}{3}$$

$$\Rightarrow \boxed{y = -w \frac{l^3}{3 EI}}$$